Week 5

- Relations

Bonus last week

Let us recall some definitions about set theory. In the following consider the sets $A = \{1, 2\}$, $B = \{3, a\}$, $C = \{1, 2, 3, a\}$.

(a) What does it mean for a set S to be a *subset* of a set T? Which of the above sets A, B and C are subsets of each other?

Solution: The set S is a subset of the set T, denoted $S \subseteq T$, if every element of S is also an element of T.

$$S \subseteq T \overset{\text{def}}{\iff} \forall x(x \in S \to x \in T)$$

We have $A \subseteq C$ and $B \subseteq C$.

{ x∈S, x∈T }

(b) What is the *power set* of a set S? What is the power set of A?

Solution: The *power set* of a set S, denoted $\mathcal{P}(S)$, is the set of all subsets of S.

$$\mathcal{P}(S) \overset{\text{def}}{=} \{T | T \subseteq S\}$$

2. (1 point) ($\star$)

Consider the set $A = \{1, 2, \emptyset\}$. List all elements of the set $S = \{A \cup \{\emptyset\}, A \cap \{\emptyset\}\} \times \{A \times \emptyset\}$ as well as of $\mathcal{P}(S)$. Your listed elements should not contain the operators $\cup, \cap$ or $\times$. You can write A or $\{1, 2, \emptyset\}$ according to your preference. Briefly explain how you arrived at your solution.

Solution:

$$S = \{(A, \emptyset), (\{\emptyset\}, \emptyset)\}$$

$$\mathcal{P}(S) = \{\emptyset, \{(A, \emptyset)\}, \{(\{\emptyset\}, \emptyset)\}, S\}.$$

{ϕ, {A, ϕ}, {{ϕ}, ϕ}, S} ✗

{ϕ, (A, ϕ), ({ϕ}, ϕ), S} ✗

Expectation. Your proof should be a sequence of logical equivalences. Each step should consist of applying the definition of $\cup, \cap, \setminus, \subseteq$ or $\emptyset$ respectively, one of the logic rules given in Lemma 2.1 of the lecture notes[1], or one of the following rules: $(F \vee G) \wedge ((H \vee G) \wedge (H \vee \neg F)) \equiv (F \vee G) \wedge (H \vee \neg F)$ (consensus theorem), $F \to G \equiv \neg F \vee G$, $F \wedge \neg F \equiv \bot$, $F \wedge \bot \equiv \bot$, $F \vee \bot \equiv F$, $F \vee \neg F \equiv \top$, $F \wedge \top \equiv F$, and $F \vee \top \equiv \top$. For this exercise, associativity is to be applied as in Lemma 2.1 *3)*. Each step of your proof should apply only a **single** rule and state **which** rule was applied. In this exercise, you may **apply the same rule multiple times** within a single proof step. Remember that applying commutativity requires a proof step.

$$
\begin{aligned}
& x \in (B \setminus C) \cup (A \setminus B) \\
\dot{\iff}\ & x \in (B \setminus C) \vee x \in (A \setminus B) && \text{(def. of } \cup\text{)} \\
\dot{\iff}\ & (x \in B \wedge \neg(x \in C)) \vee (x \in A \wedge \neg(x \in B)) && \text{(def. of } \setminus\text{)} \\
\dot{\iff}\ & \big((x \in B \wedge \neg(x \in C)) \vee x \in A\big) \wedge \big((x \in B \wedge \neg(x \in C)) \vee \neg(x \in B)\big) && \text{(distr. law)} \\
\dot{\iff}\ & \big(x \in A \vee (x \in B \wedge \neg(x \in C))\big) \wedge \big(\neg(x \in B) \vee (x \in B \wedge \neg(x \in C))\big) && \text{(comm. of } \vee\text{)} \\
\dot{\iff}\ & \Big(\big(x \in A \vee x \in B\big) \wedge \big(x \in A \vee \neg(x \in C)\big)\Big) \\
& \quad \wedge \Big(\big(\neg(x \in B) \vee x \in B\big) \wedge \big(\neg(x \in B) \vee \neg(x \in C)\big)\Big) && \text{(distr. law)} \\
\dot{\iff}\ & \Big(\big(x \in B \vee x \in A\big) \wedge \big(\neg(x \in C) \vee x \in A\big)\Big) \\
& \quad \wedge \Big(\big(x \in B \vee \neg(x \in B)\big) \wedge \big(\neg(x \in C) \vee \neg(x \in B)\big)\Big) && \text{(comm. of } \vee\text{)} \\
\dot{\iff}\ & \Big(\big(x \in B \vee x \in A\big) \wedge \big(\neg(x \in C) \vee x \in A\big)\Big) \\
& \quad \wedge \Big(\top \wedge \big(\neg(x \in C) \vee \neg(x \in B)\big)\Big) && (F \vee \neg F \equiv \top) \\
\dot{\iff}\ & \Big(\big(x \in B \vee x \in A\big) \wedge \big(\neg(x \in C) \vee x \in A\big)\Big) \\
& \quad \wedge \Big(\big(\neg(x \in C) \vee \neg(x \in B)\big) \wedge \top\Big) && \text{(comm. of } \wedge\text{)} \\
\dot{\iff}\ & \Big(\big(x \in B \vee x \in A\big) \wedge \big(\neg(x \in C) \vee x \in A\big)\Big) \wedge \big(\neg(x \in C) \vee \neg(x \in B)\big) && (F \wedge \top \equiv F) \\
\dot{\iff}\ & \big(x \in B \vee x \in A\big) \wedge \Big(\big(\neg(x \in C) \vee x \in A\big) \wedge \big(\neg(x \in C) \vee \neg(x \in B)\big)\Big) && \text{(assoc.)} \\
\dot{\iff}\ & \big(x \in B \vee x \in A\big) \wedge \big(\neg(x \in C) \vee \neg(x \in B)\big) && \text{(cons. thm.)} \\
\dot{\iff}\ & \big(x \in B \vee x \in A\big) \wedge \neg\big((x \in C) \wedge (x \in B)\big) && \text{(de Morgan)} \\
\dot{\iff}\ & x \in B \cup A \wedge \neg\big((x \in C) \wedge (x \in B)\big) && \text{(def. of } \cup\text{)} \\
\dot{\iff}\ & x \in B \cup A \wedge \neg(x \in C \cap B) && \text{(def. of } \cap\text{)} \\
\dot{\iff}\ & x \in (B \cup A) \setminus (C \cap B) && \text{(def. of } \setminus\text{)}
\end{aligned}
$$

2 steps!

2 steps!

5) $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$ *(first distributive law);*
6) $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$ *(second distributive law);*

Intuition:

$(F \vee G) \wedge ((H \vee G) \wedge (H \vee \neg F)) \equiv (F \vee G) \wedge (H \vee \neg F)$ (consensus theorem),

$H \vee G$

If RHS holds, then $H \vee G$ must be true.

$\Rightarrow$ $H \vee G$ is redundant.

Relations

Definition 3.9. A *(binary) relation* ρ *from a set* A *to a set* B (also called an (A, B)-*relation*) is a subset of $A \times B$. If $A = B$, then ρ is called a relation *on* A.

$A = \{1, 2, 3\}$ $\quad B = \{a, b, c\}$

Relation from A to B $\quad \rho = \{(1, b), (2, c)\}$

Relation on A $\quad \sigma = \{(1, 1), (1, 2), (1, 3)\}$

Relation on $\mathbb{N}$ $\quad \geq$: "greater than or equal to" relation $\quad (5, 3) \in \geq, \ (10, 10) \in \geq$

Relations are sets: $\quad < \subseteq \leq \qquad < \cap > = \phi$

Operations:

Inverse:

Definition 3.11. The *inverse* of a relation ρ from A to B is the relation $\widehat{\rho}$ from B to A defined by

$$\widehat{\rho} \stackrel{\text{def}}{=} \{(b, a) \mid (a, b) \in \rho\}.$$

$\widehat{<} \ = >$

$\widehat{\leq} \ = \geq$ $\quad$ e.g. $(1, 2) \in \leq, \ (2, 1) \in \geq$

Composition:

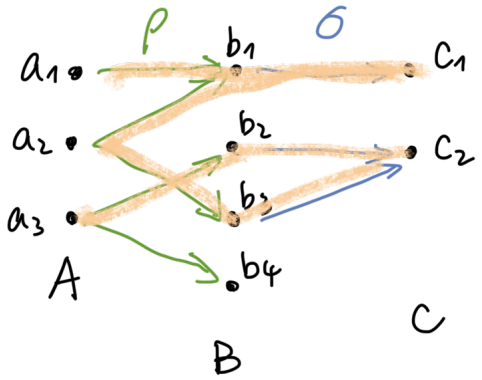
$\rho : A \text{ to } B, \quad \sigma : B \text{ to } C \Rightarrow \rho \circ \sigma : A \text{ to } C$

$$\rho \circ \sigma \stackrel{\text{def}}{=} \{(a, c) \mid \exists b \, ((a, b) \in \rho \wedge (b, c) \in \sigma)\}.$$

The n-fold composition of a relation ρ on a set A with itself is denoted ρ^n.

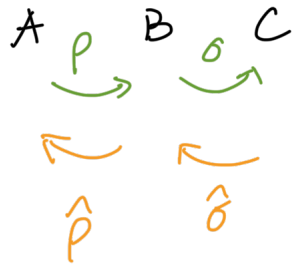
$\rho^n = \rho \circ \rho \circ \cdots \circ \rho$ (n times)

Lemma 3.7. *The composition of relations is associative, i.e., we have* $\rho \circ (\sigma \circ \phi) = (\rho \circ \sigma) \circ \phi$.



$\rho \circ \sigma = \{(a_1, c_1), (a_2, c_1), (a_3, c_2), (a_2, c_2)\}$

Lemma 3.8. *Let ρ be a relation from A to B and let σ be a relation from B to C. Then the inverse $\widehat{\rho\sigma}$ of $\rho\sigma$ is the relation $\hat{\sigma}\hat{\rho}$.*



Equivalence Relations

ρ: Relation on set A

Reflexive: $a \rho a$ for all $a \in A$ $(id \subseteq \rho)$

e.g. $A = \{a, b, c\}$

✗ $\rho = \{(a,a), (b,b)\}$ ✗

$\rho = \{(a,a), (b,b), (c,c), (a,b)\}$ ✓

irreflexive: $a \not\rho a$ for all $a \in A$

$\rho = \{(a,b), (b,c)\}$ ✓

Symmetric: $a \rho b \Leftrightarrow b \rho a$

$(\rho = \hat{\rho})$ for all $a, b \in A$

$\rho = \{(b,c), (c,b)\}$ is symmetric

"married to" is symmetric

antisymmetric: $a \rho b \wedge b \rho a \Rightarrow a = b$ for all $a, b \in A$



$\leq$ ✓ $<$ ✓

$3|-3 \wedge -3|3$ | (division) on $\mathbb{Z}$ ✗. why?

transitive $a \rho b \wedge b \rho c \Rightarrow a \rho c$ for all $a, b, c \in A$

Lemma 3.9. *A relation ρ is transitive if and only if $\rho^2 \subseteq \rho$.*

$\{(1,2), (2,3), (3,4)\}$ ✗

$\{(1,2), (2,3), (1,3)\}$ ✓

transitive closure

$$\rho^* = \bigcup_{n \in \mathbb{N} \setminus \{0\}} \rho^n = \rho^1 \cup \rho^2 \cup \rho^3 \cup \cdots$$

(a,a), (c,c), (c,b)

3.) We consider the relation $\rho = \{(a,b),(a,c),(c,a)\}$ on the set $\{a,b,c\}$. List the elements of

a) $\widehat{\rho}$: (b,a), (c,a), (a,c) *(1 Point)*

b) ρ^2 : (a,a), (c,c), (c,b) *(1 Point)*

c) ρ^* : (a,b), (a,c), (c,a), (a,a), (c,c), (c,b) *(1 Point)*

$\rho^3 = \{(a,b),(a,c),(c,a)\}$

Equivalence Relations

reflexive, symmetric, transitive

$\equiv_m$ on $\mathbb{Z}$

reflexive: $x \equiv_m x$ $\forall x$

symmetric: $x \equiv_m y \Leftrightarrow y \equiv_m x$ $\forall x\, \forall y$

transitive: For all arbitrary a,b,c

$a \equiv_m b \wedge b \equiv_m c$ $\forall a \forall b \forall c$

$\Rightarrow a - b = km \wedge b - c = \ell m,\ \ell, k \in \mathbb{Z}$

$\Rightarrow a - c = (k+\ell)m$

$\Rightarrow a \equiv_m c$ $(k+\ell \in \mathbb{Z})$

Definition 3.20. For an equivalence relation θ on a set A and for $a \in A$, the set of elements of A that are equivalent to a is called the *equivalence class of* a and is denoted as $[a]_\theta$:[22]

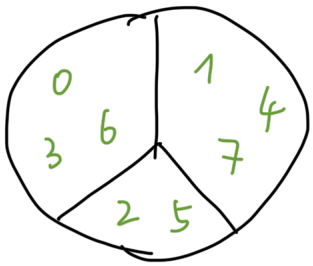
$$[a]_\theta \stackrel{\text{def}}{=} \{b \in A \mid b\ \theta\ a\}.$$

Example 3.32. The equivalence classes of the relation $\equiv_3$ are the sets

$$[0] = \{\ldots, -6, -3, 0, 3, 6, \ldots\},$$
$$[1] = \{\ldots, -5, -2, 1, 4, 7, \ldots\},$$
$$[2] = \{\ldots, -4, -1, 2, 5, 8, \ldots\}.$$

$[0]_{\equiv_3}$

Theorem 3.11. *The set A/θ of equivalence classes of an equivalence relation θ on A is a partition of A.*



ρ, σ equ. relations

$\forall x \in \rho \cap \sigma \Leftrightarrow \forall x (x \in \rho \wedge x \in \sigma)$

$\Leftrightarrow \forall x (x \rho x \wedge x \sigma x)$

① refl.

$\Leftrightarrow \forall x\ (x,x) \in \rho$

$(x,x) \in \sigma$

② symm.

③ trans.

$\Leftrightarrow \forall x\ (x,x) \in \rho \cap \sigma$

Lemma 3.10. *The intersection of two equivalence relations (on the same set) is an equivalence relation.*

Example 3.34. The intersection of $\equiv_5$ and $\equiv_3$ is $\equiv_{15}$.

partially ordered set

Order Relations

reflexive, antisymmetric, transitive

$\preceq$ on A is a poset $(A; \preceq)$

$\leq$ on $\mathbb{R}$ ✓

$\mid$ on $\mathbb{N}\setminus\{0\}$ ✓

$\mid$ on $\mathbb{N}$? Of 0

① reflexive:

for all a in $\mathbb{N}\setminus\{0\}$ $a \mid a$.

② antisymmetric

for all a, b in $\mathbb{N}\setminus\{0\}$,

$a \mid b \wedge b \mid a \overset{?}{\Rightarrow} a = b$

$\Rightarrow b = ka \wedge a = \ell b$, $k, \ell \in \mathbb{N}$

$\Rightarrow b = k\ell b$

$\Rightarrow k\ell = 1$

$\Rightarrow k = 1 \wedge \ell = 1$

$\Rightarrow a = b$

③ transitive:

for all $a, b, c \in \mathbb{N}\setminus\{0\}$

$a \mid b \wedge b \mid c$

$\Rightarrow b = ka \wedge c = \ell b$ for $k, \ell \in \mathbb{N}$

$\Rightarrow c = \ell k a$

$\Rightarrow a \mid c$ $(\ell k \in \mathbb{N})$

Total Order: for all $a, b \in A$, $a \preceq b$ or $b \preceq a$

(all elements are comparable)

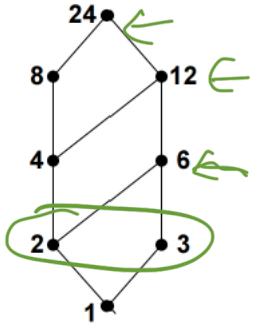
Hasse Diagrams:

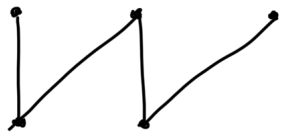
edge from a to b if b covers a ($a \prec b$ and there is no c with $a \prec c \wedge c \prec b$)

Hasse Diagram of the poset $(\{1, 2, 3, 4, 6, 8, 12, 24\}; \mid)$

Definition 3.29. Let $(A; \preceq)$ be a poset, and let $S \subseteq A$ be some subset of A. Then

1. $a \in A$ is a *minimal (maximal) element* of A if there exists no $b \in A$ with $b \prec a$ ($b \succ a$).[32]
2. $a \in A$ is the *least (greatest) element* of A if $a \preceq b$ ($a \succeq b$) for all $b \in A$.[33]
3. $a \in A$ is a *lower (upper) bound*[34] of S if $a \preceq b$ ($a \succeq b$) for all $b \in S$.[35]
4. $a \in A$ is the *greatest lower bound (least upper bound)* of S if a is the greatest (least) element of the set of all lower (upper) bounds of S.[36]





Definition 3.31. Let $(A; \preceq)$ be a poset. If a and b (i.e., the set $\{a, b\} \subseteq A$) have a greatest lower bound, then it is called the *meet* of a and b, often denoted $a \wedge b$. If a and b have a least upper bound, then it is called the *join* of a and b, often denoted $a \vee b$.

Definition 3.32. A poset $(A; \preceq)$ in which every pair of elements has a meet and a join is called a *lattice*[39].

glb

lub

Definition 3.30. A poset $(A; \preceq)$ is *well-ordered*[37] if it is totally ordered and if every non-empty subset of A has a least element.[38]

$(\mathbb{N}; \leq)$: well-ordered

$(\mathbb{Z}; \leq)$: not well-ordered $\{\dots, -3, -2, -1\}$

Definition 3.28. The *direct product* of posets $(A; \preceq)$ and $(B; \sqsubseteq)$, denoted $(A; \preceq) \times (B; \sqsubseteq)$, is the set $A \times B$ with the relation $\leq$ (on $A \times B$) defined by

$$(a_1, b_1) \leq (a_2, b_2) \overset{\text{def}}{\iff} a_1 \preceq a_2 \wedge b_1 \sqsubseteq b_2.$$

Theorem 3.12. *$(A; \preceq) \times (B; \sqsubseteq)$ is a partially ordered set.*

Theorem 3.13. *For given posets $(A; \preceq)$ and $(B; \sqsubseteq)$, the relation $\leq_{\text{lex}}$ defined on $A \times B$ by*

$$(a_1, b_1) \leq_{\text{lex}} (a_2, b_2) \overset{\text{def}}{\iff} a_1 \prec a_2 \vee (a_1 = a_2 \wedge b_1 \sqsubseteq b_2)$$

is a partial order relation.[31]

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Meet: greatest lower bound

Join: least upper bound

Definition 3.32. A poset $(A; \preceq)$ in which every pair of elements has a meet and a join is called a *lattice*[39].